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A Novel Foundation Model-Based Framework for Multimodal Retinal Age Prediction

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ABSTRACT The retinal age gap (RAG; the difference between the retina's biological and chronological age) has recently gained increased attention as a potential image-based, non-invasive, and accessible biomarker for a broad spectrum of ocular and non-ocular diseases. Traditionally, machine learning predictions of biological retinal age utilize convolutional neural network (CNN) architectures and data from color fundus photography (CFP). Despite being previously unexplored, the multimodal fusion of two-dimensional CFP with three-dimensional optical coherence tomography (OCT) data has significant potential to enhance retinal age prediction accuracy and the diagnostic utility of the RAG biomarker. Therefore, this work presents a novel foundation model-based framework for multimodal retinal age prediction. Technology or Method: Feature representations from CFP and OCT images were extracted using RETFound, a powerful foundation model for retinal image analysis. These representations were then combined using an innovative fusion strategy to train a lightweight linear regression head model for predicting retinal age. Training and evaluation of the developed multimodal retinal age prediction model was achieved using retinal images from over 80,000 participants in the UK Biobank. Results: The developed multimodal model sets a new benchmark in retinal age prediction (mean absolute error of 2.75 years), outperforming traditional CNN and single-modality approaches. Additionally, multimodal RAG values demonstrated superior performance in classifying patients with diabetes mellitus type 1, multiple sclerosis, and chronic kidney disease, highlighting the clinical relevance of the proposed multimodal approach for non-ocular disease detection. Conclusions: This work demonstrates that multimodal fusion of CFP and OCT significantly improves retinal age prediction and subsequent RAG-based analyses. By leveraging foundation models and multimodal retinal imaging, the proposed approach enhances disease classification accuracy and demonstrates the potential of integrating the RAG into clinical workflows as a scalable, non-invasive screening tool. Significance: The findings underscore the potential of multimodal retinal imaging to transform RAG into a clinically relevant and highly accessible biomarker for disease detection.

INDEX TERMS Foundation model, machine learning, retinal imaging, retinal age prediction.

Clinical Impact—The methods developed in this work demonstrate strong translational potential to enhance the detection of multiple diseases based on multi-modal retina images in clinical settings. The proposed framework provides a non-invasive and accessible screening tool that holds promise for improving diagnostic capabilities and patient outcomes across global healthcare practices.

I. INTRODUCTION

THE retinal age gap (RAG; the difference between the retina's biological and chronological age) has recently

gained attention as a potential image-based biomarker with high sensitivity for several diseases, such as renal failure [1], stroke [2], and Parkinson's disease [3], and may also

be predictive of the presence of several others. One of the major advantages of using the RAG as a biomarker lies in its ability to be measured using standard, low-cost eye imaging equipment. Such imaging technology is commonly available in ophthalmic practices around the world and is even adaptable to at-home and remote data collection [4]. The accessibility and non-invasiveness of the RAG assessment positions it as a particularly valuable tool for large-scale, global disease screening. Practically, significant RAG deviations may be used by clinicians to identify high-risk individuals who may benefit from further diagnostic evaluation, potentially enabling earlier disease detection and timely therapeutic intervention.

The RAG is typically computed by measuring the difference between an individual's true chronological age and the biological age of their retina, which is commonly estimated using a machine learning model that was trained on retinal data from healthy subjects for which the biological age is assumed to match the chronological age. Consequently, improving the performance of retinal age prediction models is crucial as it directly impacts the reliability and sensitivity of the RAG as a biomarker for various diseases. Traditional approaches for predicting the biological retinal age have predominantly relied on standard convolutional neural network (CNN) architectures, with the Xception CNN architecture being used in most cases [1], [2], [3], [5], [6], [7]. Additionally, Nielsen et al. have recently shown that retinal age prediction models can be trained based on color fundus photography (CFP) using distributed learning methods [8].

Within the field of retinal imaging, CFP and optical coherence tomography (OCT) are the two most prevalent, non-invasive imaging modalities, each with distinct features that enable a comprehensive assessment of retinal health [9]. CFP provides two-dimensional (2D) color images of the retina [10]. In particular, CFP captures detailed images that display the optic disc, macula, and retinal vasculature, allowing for the visualization and calculation of retinal features related to the blood vessel architecture, optic disc morphology, and the presence of retinal lesions or other abnormalities [10], [11], [12]. In contrast, OCT captures three-dimensional cross-sectional images of the retina and surrounding structures, allowing for high-resolution visualization and precise measurement of retinal layer thickness at the micrometer scale [14]. OCT provides critical insights into the presence of fluid accumulation, the integrity of the retinal layers, and the detailed morphology of the macula and optic nerve head [14], [15], [16].

CFP has traditionally been the primary imaging source used to train ML models for retinal age prediction. In contrast, OCT has rarely been used for this purpose directly although it has been shown to capture structural changes within the retina that are associated with aging [1], [3], [5], [6], [7], [17]. Notably, it has been observed that the thickness of both, the macula and the retinal nerve fiber layer, tends to decrease as individuals age [18], [19], [20]. Furthermore, age-related decline in vascular attributes, including blood

flow velocity, which may influence vessel thickness, and retinal vessel density, have also been documented [21], [22]. Expanding upon these observations, a study conducted by Chen et al. [24] revealed that the gap between an individual's chronological age and the biological age predicted by OCT-derived biomarkers might indicate an increased risk of mortality.

More recently, various studies, not specific to retinal age prediction, have shown that combining CFP and OCT imaging data can significantly improve the performance of ML models for ocular disease classification tasks [25]. For example, research by He et al. [26] and Lim et al. [27] suggest that integrating modality-specific features from both, CFP and OCT, can increase the accuracy of age-related macular degeneration and glaucoma diagnosis. Moreover, the combination of multimodal imaging data mirrors actual clinical settings, where ophthalmologists may examine both CFP and OCT images concurrently, utilizing the unique information each modality offers to achieve more precise diagnoses [27], [28]. However, multimodal fusion has not been investigated for retinal age prediction so far, although it has the potential to improve model performance and enhance the diagnostic capability of the RAG as a disease biomarker.

Thus, the aim of this work was to develop and test the first multimodal approach for biological retinal age prediction through fusion of CFP and OCT imaging data. Therefore, the proposed, novel method utilizes the advanced feature extraction capabilities of RETFound, a large foundation model specifically designed for the analysis of CFP and OCT images, which underwent pretraining through self-supervised representation learning methods using a large database of unlabeled images [30]. The overall aim of our study is to demonstrate that state-of-the-art retinal age prediction performance can be achieved by using a RETFound-based linear head (RLH) model that combines imaging features derived using RETFound from both CFP and OCT images. Importantly, the proposed method provides a lightweight, highly efficient approach for foundation model adaptation in this multimodal setting. This approach has the potential to be generalized for a wide variety of downstream prediction tasks. Additionally, the method's computational and data efficiency makes it well-suited for use in regions with limited network and computational resources. Furthermore, we aimed to investigate to what extent the combined use of CFP and OCT data through the proposed approach for multimodal retinal age prediction enhances the predictive performance beyond what is achievable when only a single imaging modality is used.

The key contributions of this work can be summarized as follows: (1) To the best of our knowledge, we present the first multimodal retinal age prediction model that combines CFP and OCT data. Additionally, we are the first to utilize the RETFound foundation model for multi-modal retinal age prediction and compare the performance to standard CNN architectures. (2) We conduct a comprehensive evaluation of the proposed model by leveraging CFP and OCT imaging

data from over 80,000 individuals from the UK Biobank. (3) We explore how the amount of available training data influences the retinal age prediction performance of the model and compare the results to Xception, a single-modality CNN architecture commonly used for retinal age prediction. (4) We explore if the multimodal RAG predictions can be used to detect a range of relevant health conditions, including systemic diseases known to be associated with changes of the retina, such as diabetes mellitus type 1, multiple sclerosis, and chronic kidney disease. Furthermore, we assess whether the improved accuracy in retinal age estimation achieved through multimodal fusion of CFP and OCT translates into enhanced downstream disease detection capabilities.

II. MATERIALS AND METHODS

A. DATASET

This work leverages baseline data from the UK Biobank, a comprehensive, population-based longitudinal study designed to investigate the genetic and environmental determinants affecting diseases in individuals of middle to old age [31]. Over 500,000 participants aged 40 to 69 years were recruited to the UK Biobank from across England, Scotland, and Wales between 2006 and 2010, focusing on gathering detailed baseline data for comprehensive health-related outcomes analysis. Therefore, participants underwent comprehensive baseline assessments, including digital healthcare questionnaires, medical imaging, extensive physical exams, and collection of clinical records utilizing the 10th Revision of the International Statistical Classification of Diseases and Related Health Problems (ICD-10) codes. The UK Biobank adheres to the Declaration of Helsinki and received approval from the National Information Governance Board for Health and Social Care, and the NHS North West Multicenter Research Ethics Committee.

The UK Biobank features a substantial collection of CFP and OCT images, with ophthalmic examinations carried out using the Topcon 3D OCT-1000 Mark II scanner [32]. CFP and OCT images were concurrently acquired from over 80,000 participants, under nonmydriatic conditions. Consequently, for each participant, a matching OCT image was acquired together with each CFP image, within the same acquisition session. The OCT imaging protocol entailed scanning a 6 mm × 6 mm region of the retinal layers surrounding the macula, capturing 128 B-scans. Each B-scan included 512 A-scans, with a total acquisition time of 3.7 seconds. CFP images were obtained with a digital camera integrated into the 3D OCT-1000 system, which took images of the posterior pole centered on the macula while also encompassing the optic disc.

B. DATA PREPROCESSING

Within the UK Biobank database, a total of 177,354 matched CFP and OCT images were collected from 86,522 participants. To ensure compatibility with the RETFound model, the original CFP images (2048 × 1536 pixels) were preprocessed by centering the retina within the frame and applying a center

crop to create a square format while preserving the maximum retinal area. The cropped images were then uniformly resized to 224 × 224 pixels, maintaining the original aspect ratio to prevent any unwanted distortion. Similarly, the OCT images initially captured with a size of 712 × 512 × 128 voxels were center cropped to a size of 512 × 512 × 128 and then uniformly resampled to a size of 224 × 224 × 128 voxels. Moreover, as the RETFound model was pretrained specifically and exclusively for middle B-scans [30], only the central 20 B-scans from each OCT image were selected and used for subsequent analysis in this work. This selection ensures consistency with the pretrained model and prevents the inclusion of data that RETFound was not trained or designed to process, thereby preserving the integrity of feature extraction and model performance. Quality assessments were performed on CFP and OCT images to remove those affected by motion artifacts and other issues related to their acquisition. Only participants with matched CFP and OCT images after quality control were used for further analysis. The quality evaluation of CFP images employed a deep learning-based method proposed by Fu et al. [33], while quality control for the OCT images was based on the image quality score computed by the Topcon Advanced Boundary Segmentation (TABS) algorithm as described by Chen et al. [24]. This process led to the selection of 92,091 high-quality matched CFP/OCT images from 49,369 participants that met the quality standards and were chosen for further experimental analysis in this study.

C. PROPOSED RETINAL AGE PREDICTION MODEL

The RETFound model [30] leverages a masked autoencoder framework, which is divided into two main modules: an encoder and a decoder. However, only the encoder module was used for feature extraction in this work. The encoder utilizes a vision transformer large-scale (ViT-large) architecture, comprising of 24 transformer blocks [34]. This framework processes each 16 × 16 pixel patch of the input retinal images (either 2D CFP image or 2D OCT slice) to produce feature representations for each patch. Each block within this setup includes a multilayer perceptron and a multi-headed self-attention mechanism. Zhou et al. [30] publicly released two pretrained versions of the RETFound model: one optimized for CFP data and the other trained using OCT images. The CFP-specific model was pretrained on a very large database of 904,170 unlabeled retinal CFP images from the MEH-MIDAS [30] and Kaggle EyePACS datasets [35], while the OCT-specific model was trained on 736,442 unlabeled OCT images from MEH-MIDAS [30] and a dataset curated by Kermany et al. [36]. The training process for both models employed a self-supervised learning strategy known as masked image completion. This innovative approach involves intentionally obscuring parts of the input images during training, tasking the model to learn to predict the occluded imaging information. This method to train foundation models is expected to not only enhance the model's ability to understand and interpret complex visual

patterns but also significantly improves its generalizability and effectiveness in processing and analyzing new, unseen CFP and OCT images.

In our proposed method, each CFP image from the UK Biobank dataset is processed using a forward pass through the frozen CFP-RETFound encoder, resulting in the creation of a 1,024-dimensional feature representation for each CFP image. Likewise, for each associated OCT image, the central 20 slices are separately processed via the frozen OCT-RETFound encoder, producing a 1,024-dimensional feature representation for every slice. The feature representations from each slice were then combined to create a single 1,024-dimensional feature representation for every OCT image using global average pooling [37].

For the experiments, we evaluated three distinct linear regression head (LRH) models using the extracted feature representations. The use of the term “head” in this context indicates that only the final linear layer is actively trained for the task of predicting retinal age, with the parameters of the RETFound models, which are responsible for generating these representations, remaining fixed. The first model utilized only the CFP feature representations for training, the second model solely makes use of the OCT feature representations, and the third model underwent training using a combined multimodal concatenation of both CFP and OCT feature representations. LRH models were trained to minimize the ridge regression loss between the true chronological age and the predicted age, using an L2 regularization constant of five. These models were developed using scikit-learn 1.0.2 and the training process was carried out on an Intel i7-11700KF CPU. Fig. 1 provides a graphical representation of the established workflow.

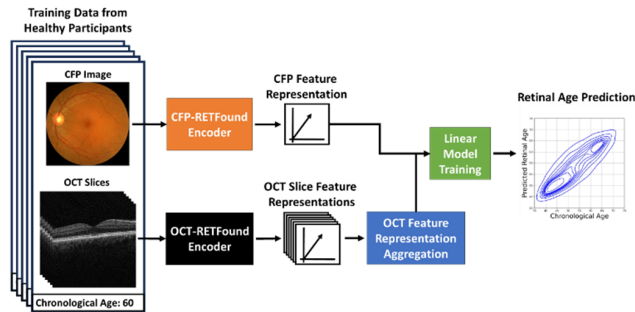


FIGURE 1. Proposed framework for estimating biological retinal age employing a linear regression head model, trained on feature representations extracted from CFP and OCT images using the frozen CFP-RETFound and OCT-RETFound foundation model encoders, respectively.

D. BASELINE RETINAL AGE PREDICTION MODEL

Several previous studies investigating the potential of the RAG as a disease biomarker have employed the Xception CNN architecture [38] for predicting the biological retinal age using CFP data, making it an especially relevant baseline for comparison [1], [2], [3], [5], [6], [7]. Consequently, in this work, we compared our method against the Xception

model [39], which has about 23 million parameters across 71 layers, and was pretrained using ImageNet, which contains diverse natural images. The Xception model was fine-tuned on the CFP data with the Adam optimizer, applying a learning rate of 1×10^{-3} and a weight decay of 1×10^{-5} to minimize the L2 loss between the actual chronological age and its predicted counterpart. Training was performed for 200 epochs with a batch size of 16. To prevent overfitting, model selection was performed by selecting the training iteration with optimal validation performance. The Xception model was implemented within PyTorch 1.13.1 and training was performed using an NVIDIA RTX 3090 GPU.

E. DATASET CONSTRUCTION AND COMPUTATION OF RAG

Healthy participants for training the age prediction models were selected based on the criteria established by Zhu et al. [6], using only images of participants without any self-reported medical conditions prior to imaging. Overall, CFP and OCT images from both eyes of 7,138 healthy participants were used for model training and evaluation, with detailed demographics presented in Table 1. An 80%/10%/10% participant-level split for training, testing, and validation was applied across ten Monte-Carlo cross-validation iterations, using two images per participant (one from each eye). Supplementary Tables 1 and 2 contain the demographic information for the training and test datasets, respectively. Additionally, we performed statistical comparisons between the cohorts, demonstrating that there are no significant differences in these variables, and ensuring that the hold-out test set is representative of the broader dataset and is suitable to allow for fair model evaluation. To assess the effect of training data volume on model performance, training data was partitioned into subsets of 150, 300, 600, 1200, 2400, and 4800 participants, enabling a detailed performance analysis across varying training data sizes. To prevent statistical bias, CFP and OCT images were carefully divided, ensuring that images from the same participant were not mixed across the training, testing, and validation sets, thereby avoiding cross-contamination.

TABLE 1. Demographic information of the 7,138 healthy participants included for training and evaluating the retinal age prediction models. N: number of participants with available demographic information in UK Biobank; Std: standard deviation; Min: minimum observed value; Q1: quartile 1; Q2: quartile 2; Q3: quartile 3; Max: maximum observed value; LogMAR: logarithm of the minimum angle of resolution.

	N	Mean	Std	Min	Q1	Q2	Q3	Max
Age (years)	7,138	52.21	7.94	40	45	51	59	74
Diastolic blood pressure (mmHg)	7,118	80.18	9.67	45.00	73.50	79.50	86.50	118.50
Systolic blood pressure (mmHg)	7,118	131.83	16.90	90.00	119.50	130.00	142.50	226.50
HDL cholesterol (mmol/L)	6,219	1.53	0.39	0.53	1.25	1.48	1.76	3.41
Triglycerides (mmol/L)	6,548	1.53	0.93	0.25	0.92	1.27	1.84	11.01
Glucose (mmol/L)	6,208	5.00	0.60	2.91	4.67	4.95	5.26	16.30
Weight (kg)	7,111	75.28	14.65	40.30	64.40	73.80	84.60	166.60
Body mass index (kg/m ²)	7,110	26.09	4.09	15.52	23.27	25.62	28.36	51.34
Arterial stiffness index	7,060	8.90	4.90	2.04	6.75	8.42	10.53	350.00
Left eye logMAR	7,117	-0.00	0.20	-0.48	-0.14	-0.06	0.06	1.35
Right eye logMAR	7,130	-0.00	0.20	-0.44	-0.12	-0.06	0.06	1.35
Alcohol intake	7,119	Never: 449, Special occasions only: 775, One to three times a month: 801, Once or twice a week: 1,919, Three or four times a week: 1,753, Daily or almost daily: 1,412, Prefer not to answer: 10						
Smoking status	7,119	Never smoked: 4,297, Previous smoker: 2,080, Current smoker: 720, Prefer not to answer: 22						
Sex	7,138	Female: 3,915, Male: 3,223						
Ethnicity	7,119	White: 6,497, Mixed background: 72, Black: 178, Asian: 190, Other: 182						

To assess the RAG's effectiveness as a disease biomarker across the trained models, RAG values were calculated for participants with a self-reported disease at any point prior to imaging, by subtracting the chronological age from the predicted biological age. Biological age prediction models are often subject to age-related biases, typically manifesting as an overestimation of age in younger individuals and an underestimation in older individuals [40]. To correct for age distribution discrepancies between the training group and disease groups, de Lange et al.'s bias correction methods, originally developed for brain age prediction models, were applied to correct the RAG values [41]. To explore the relationship between RAG values and various disease groups, we computed RAG scores, which represent the average of the age-corrected RAG values for each disease group. As such, healthy individuals are expected to have a RAG score of zero. The magnitude of the RAG scores serves as a measure to quantify the extent to which disease groups deviate from the baseline established by healthy controls.

To reduce bias, only one image per participant was analyzed for 39,254 participants with a self-reported disease, prioritizing the right eye, and opting for the left eye if the right was not available [1]. Table 2 lists demographic details of this cohort. A summary of the data utilized for both training the age prediction models and for analyzing the disease categories is depicted in Fig. 2. To explore RAG value discrepancies in patients with diseases with known retinal pathologies, we analyzed data from participants with diabetes mellitus type 1 (169 participants), multiple sclerosis (81 participants), and chronic kidney disease (411 participants), grouped by ICD-10 codes acquired prior to imaging (detailed in Supplementary Table 3). We analyzed RAG values across these groups to assess statistical differences in the trained models. Moreover, we investigated whether the RAG value could serve as a continuous risk score for disease classification. For this analysis, we computed RAG values for both healthy participants in the test set and for participants diagnosed with diabetes mellitus type 1, multiple sclerosis, or chronic kidney disease. We then evaluated the ability of these RAG values to distinguish between healthy individuals and individuals with disease by calculating the area under the receiver operating characteristic curve (AUROC). This AUROC-based assessment provided a quantitative measure of the RAG value's effectiveness as a biomarker for disease detection.

F. PERFORMANCE EVALUATION

We assessed the computational performance of the models using the caltflops Python library [42] to determine the total floating-point operations (FLOPs) required for training. The relationship between predicted biological and chronological ages in healthy participants was analyzed using Pearson's correlation coefficient and mean absolute error (MAE), with a strong correlation defined as a Pearson coefficient with an absolute value above 0.7 [43]. Disease classification performance was evaluated using the AUROC.

TABLE 2. Demographic information of the 39,254 participants with self-reported disease prior to image acquisition. N: number of participants with available demographic information in UK Biobank; Std: standard deviation; Min: minimum observed value; Q1: quartile 1; Q2: quartile 2; Q3: quartile 3; Max: maximum observed value; LogMAR: logarithm of the minimum angle of resolution.

	N	Mean	Std	Min	Q1	Q2	Q3	Max
Age (years)	39,254	57.15	8.04	40	51	59	64	76
Diastolic blood pressure (mmHg)	39,108	81.90	10.07	44.50	75.00	81.50	88.50	134.00
Systolic blood pressure (mmHg)	39,108	137.62	18.48	76.50	124.50	136.00	149.00	239.00
HDL cholesterol (mmol/L)	33,904	1.48	0.39	0.36	1.20	1.43	1.71	4.13
Triglycerides (mmol/L)	35,864	1.69	0.96	0.25	1.04	1.45	2.07	11.19
Glucose (mmol/L)	33,874	5.16	1.06	1.25	4.71	5.01	5.35	25.49
Weight (kg)	39,093	78.35	16.04	33.60	66.70	76.70	87.90	188.10
Body mass index (kg/m ²)	39,078	27.43	4.80	12.65	24.13	26.74	29.90	65.98
Arterial stiffness index	38,887	9.62	3.41	0.97	7.34	9.35	11.35	215.00
Left eye logMAR	39,103	0.02	0.21	-1.06	-0.10	-0.04	0.10	1.35
Right eye logMAR	39,131	0.02	0.20	-0.44	-0.10	-0.04	0.10	1.35
Alcohol intake	39,199	Never: 3,083, Special occasions only: 4,574, One to three times a month: 4,560, Once or twice a week: 9,827, Three or four times a week: 9,071, Daily or almost daily: 8,046, Prefer not to answer: 38						
Smoking status	39,199	Never smoked: 21,988, Previous smoker: 13,801, Current smoker: 3,284, Prefer not to answer: 126						
Sex	39,254	Female: 21,699, Male: 17,555						
Ethnicity	39,199	White: 36,893, Mixed background: 266, Black: 719, Asian: 691, Other: 630						

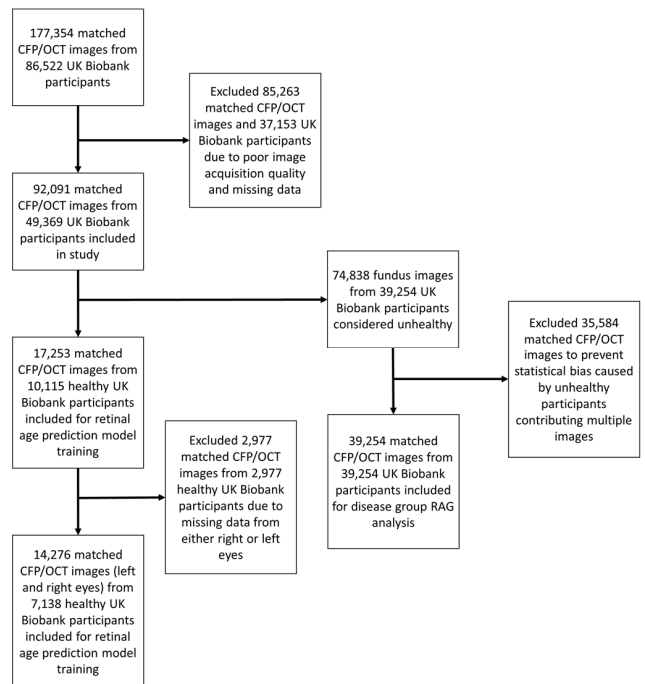


FIGURE 2. Overview of participant data used in this study, illustrating how the data was divided to train machine learning models for retinal age prediction and for analyzing the RAG in individuals with self-reported disease.

Statistical differences in demographic variables between the training and test datasets were assessed using unpaired t-tests for continuous variables and chi-squared tests for categorical variables. To assess the consistency and reliability of the RAG predictions, we measured the mean absolute difference (MAD) between the estimated retinal ages of the left and right eyes for each participant. Statistical differences in model performance were assessed with paired t-tests. The Bonferroni correction was used to adjust for multiple comparisons, taking into account the six dataset sizes and $\binom{4}{2} = 6$ pairwise comparisons among the four model types. As such, a p-value below $\alpha = \frac{0.05}{36}$ was deemed significant.

III. RESULTS

A. AGE PREDICTION PERFORMANCE

Table 3 and Table 4 present the MAEs and Pearson correlation coefficients comparing the predicted biological ages with actual chronological ages for the trained retinal age prediction models evaluated on the hold-out test set of 714 healthy participants. Additionally, Supplementary Figures 1 and 2 present boxplots of these results, and Supplementary Table 4 shows the MAE results by age. Generally, the performance of all models was found to improve with larger training datasets. For the smallest training dataset of 150 participants, the MAEs and Pearson correlation coefficients were as follows: the multimodal RLH model trained using both CFP and OCT data had an MAE of 3.52 ± 0.06 and a Pearson correlation coefficient of 0.82 ± 0.01 ; the single-modality Xception baseline model trained using CFP data achieved an MAE of 4.28 ± 0.25 and a Pearson correlation coefficient of 0.73 ± 0.03 ; the single-modality RLH model trained using only CFP data had an MAE of 3.75 ± 0.08 and a Pearson correlation coefficient of 0.79 ± 0.02 ; and the single-modality RLH model trained using only OCT data led to an MAE of 4.88 ± 0.10 and a Pearson correlation coefficient of 0.68 ± 0.02 . When the training dataset was increased to 4800 participants, the results improved substantially: the multimodal RLH model trained on both CFP and OCT data achieved an MAE of 2.75 ± 0.02 and a Pearson correlation coefficient of 0.90 ± 0.01 ; the single-modality Xception model trained using CFP data resulted in an MAE of 3.36 ± 0.06 and a Pearson correlation coefficient of 0.84 ± 0.01 ; the single-modality RLH model trained using only CFP data led to an MAE of 3.21 ± 0.01 and a Pearson correlation coefficient of 0.86 ± 0.01 ; and the single-modality RLH model trained using only OCT data achieved an MAE of 3.98 ± 0.02 and a Pearson correlation coefficient of 0.80 ± 0.01 .

The proposed multimodal RLH model, which utilizes both CFP and OCT data, consistently outperformed all other tested models (single-modality RLHs and CFP CNN) across varying training dataset sizes, providing a statistically significant gain in performance measured across the MAE ($p < 0.001$) and Pearson correlation coefficient ($p < 0.001$). In contrast, the RLH model trained exclusively on OCT data had the lowest performance in terms of the MAE ($p < 0.001$) and Pearson correlation coefficient ($p < 0.001$) across all dataset sizes. Additionally, the performance of the RLH model trained solely on CFP data significantly outperformed the baseline Xception CNN model in terms of the MAE ($p < 0.001$) and Pearson correlation coefficient ($p < 0.001$), for all training set sizes except when trained using 1200 participants, where p-values were above the Bonferroni-corrected significance threshold for MAE ($p = 0.018$) and Pearson correlation coefficient ($p = 0.013$).

Table 5 shows the MAD between the estimated retinal ages of the left and right eyes for each participant in the test set. Generally, the MAD of all RETFound-based models was found to decrease with larger training datasets, while

TABLE 3. Mean absolute error (MAE) of the trained models, comparing predicted biological ages with actual chronological ages for healthy subjects in the hold-out test set.

Training set size	Xception (CFP)	RLH (CFP)	RLH (OCT)	RLH (CFP + OCT)
150	4.28 ± 0.25	3.75 ± 0.08	4.88 ± 0.10	3.53 ± 0.06
300	3.97 ± 0.15	3.63 ± 0.06	4.59 ± 0.08	3.31 ± 0.10
600	3.70 ± 0.05	3.49 ± 0.07	4.37 ± 0.06	3.12 ± 0.08
1200	3.46 ± 0.09	3.37 ± 0.06	4.20 ± 0.04	2.96 ± 0.07
2400	3.40 ± 0.08	3.28 ± 0.03	4.08 ± 0.02	2.83 ± 0.04
4800	3.36 ± 0.06	3.21 ± 0.01	3.98 ± 0.02	2.75 ± 0.02

TABLE 4. Pearson correlation coefficient analysis of the trained models, comparing predicted biological ages with actual chronological ages for healthy subjects in the hold-out test set.

Training set size	Xception (CFP)	RLH (CFP)	RLH (OCT)	RLH (CFP + OCT)
150	0.73 ± 0.03	0.79 ± 0.02	0.68 ± 0.02	0.82 ± 0.01
300	0.77 ± 0.02	0.81 ± 0.01	0.73 ± 0.02	0.84 ± 0.01
600	0.80 ± 0.01	0.82 ± 0.01	0.75 ± 0.01	0.86 ± 0.00
1200	0.83 ± 0.01	0.84 ± 0.01	0.77 ± 0.01	0.88 ± 0.01
2400	0.84 ± 0.01	0.85 ± 0.01	0.79 ± 0.01	0.89 ± 0.00
4800	0.84 ± 0.01	0.86 ± 0.01	0.80 ± 0.01	0.90 ± 0.01

the Xception model did not show such a decrease in MAD. For the smallest training dataset (150 participants), the multimodal RLH model trained on both CFP and OCT data achieved a MAD of 2.26 ± 0.10 , outperforming the single-modality Xception baseline model trained on CFP alone (MAD = 2.41 ± 0.29). The single-modality RLH model trained on CFP alone, resulted in a MAD of 2.31 ± 0.11 , while the RLH model trained on OCT alone had a MAD of 2.37 ± 0.20 . With a larger training dataset (4,800 participants), the multimodal RLH model trained on CFP and OCT further improved, achieving the lowest MAD of 1.94 ± 0.03 . In comparison, the Xception model trained on CFP alone demonstrated a higher MAD of 2.72 ± 0.14 , highlighting its lower intra-patient eye consistency. The single-modality RLH model trained on CFP achieved a MAD of 2.13 ± 0.02 , while the single-modality RLH model trained on OCT alone resulted in a MAD of 2.20 ± 0.02 . Importantly, when trained using data from 4800 participants, all RLH-based models (CFP, OCT, and combined) demonstrated significantly better inter-eye consistency than the Xception baseline ($p < 0.001$ for all comparisons using paired t-test).

TABLE 5. mean absolute difference (MAD) between the estimated retinal ages of the left and right eyes for each healthy subjects in the hold-out test set.

Training set size	Xception (CFP)	RLH (CFP)	RLH (OCT)	RLH (CFP + OCT)
150	2.41 ± 0.29	2.31 ± 0.11	2.37 ± 0.20	2.26 ± 0.10
300	2.48 ± 0.35	2.36 ± 0.10	2.38 ± 0.14	2.28 ± 0.08
600	2.32 ± 0.20	2.33 ± 0.12	2.35 ± 0.11	2.21 ± 0.08
1200	2.40 ± 0.16	2.24 ± 0.06	2.28 ± 0.06	2.10 ± 0.07
2400	2.43 ± 0.09	2.15 ± 0.05	2.22 ± 0.04	1.96 ± 0.03
4800	2.72 ± 0.14	2.13 ± 0.02	2.20 ± 0.02	1.94 ± 0.03

B. RAG PREDICTION STABILITY ANALYSIS

To evaluate the effectiveness of the RAG as a disease biomarker across various models, we used the trained models to compute RAG values for 39,254 individuals who

had self-reported a disease before undergoing imaging. The standard deviation of the age-corrected RAG values was calculated for each participant to assess how consistently RAG values can be estimated for the same participants across the RLH models (CFP, OCT, CFP + OCT), and the Xception model. As illustrated in Table 6, the RAG standard deviation substantially decreases with more training data for all RLH models (CFP, OCT, and CFP + OCT), demonstrating a link between larger datasets and more stable RAG predictions. Furthermore, for datasets of 300 participants or more, the RLH models show significantly lower RAG standard deviations ($p < 0.001$) than the Xception model, suggesting that the RLH-based models provide better RAG prediction stability.

TABLE 6. Retinal age gap (RAG) standard deviation across individuals with self-reported disease.

Training set size	Xception (CFP)	RLH (CFP)	RLH (OCT)	RLH (CFP + OCT)
150	2.03 ± 0.60	2.05 ± 0.63	1.88 ± 1.09	2.13 ± 0.77
300	1.96 ± 0.56	1.81 ± 0.58	1.51 ± 0.76	1.83 ± 0.61
600	1.80 ± 0.48	1.52 ± 0.49	1.29 ± 0.70	1.56 ± 0.53
1200	1.57 ± 0.42	1.18 ± 0.40	0.99 ± 0.64	1.22 ± 0.47
2400	1.65 ± 0.43	0.85 ± 0.28	0.73 ± 0.53	0.88 ± 0.33
4800	1.81 ± 0.51	0.44 ± 0.14	0.38 ± 0.34	0.47 ± 0.19

C. RAG SCORE ANALYSIS

As previously mentioned, RAG scores are defined as the expected values of the age-corrected RAG distributions for disease groups. Therefore, healthy participants are expected to have a RAG score of zero, while the magnitude of the RAG scores acts as a metric to quantify the level of deviation of disease groups from the baseline set by healthy controls. Fig. 3 shows, for models trained using 4,800 participants, the mean RAG scores for patients with diabetes mellitus type 1 (169 participants), multiple sclerosis (81 participants), and chronic kidney disease (411 participants). All RLH models (CFP, OCT, and CFP + OCT) consistently yielded higher RAG scores than the Xception model, indicating that the RLH models provide more informative RAG estimates for these disease groups. For example, the multimodal (CFP + OCT) RLH model, when trained on data from 4800 participants, led to RAG scores of 3.59 ± 0.13 years for diabetes mellitus type 1, 2.07 ± 0.09 years for multiple sclerosis, and 1.84 ± 0.06 years for chronic kidney disease, respectively. Furthermore, the RAG scores generated by the combined CFP + OCT RLH model generally fell between the scores estimated by the CFP RLH and OCT RLH models, suggesting that information from these two imaging modalities was indeed used jointly by the multimodal model. Moreover, larger training data volumes were shown to generally increase RAG scores in all RLH models, implying that the disease detection capability improves for models with improved retinal age prediction performance.

Table 7 summarizes the AUROC scores obtained by using the RAG value as a continuous risk score to classify participants with diabetes mellitus type 1, multiple sclerosis, and chronic kidney disease. These AUROC

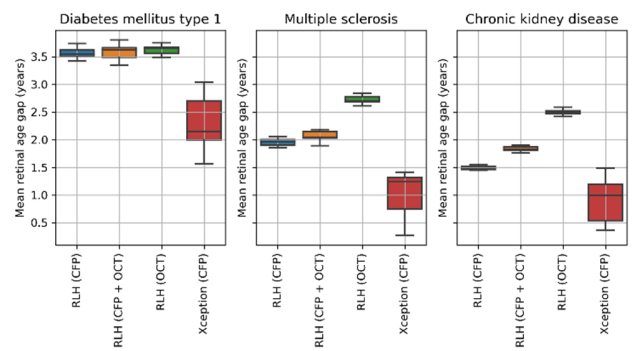


FIGURE 3. Boxplots of mean retinal age gap (RAG) scores for each model trained on 4,800 participants, evaluated across the three selected diseases of interest.

metrics were computed from RAG values estimated using the retinal age prediction models trained on data from 4,800 participants. The multimodal CFP + OCT RLH model demonstrated the highest classification performance, achieving AUROC values of 0.683 ± 0.003 for chronic kidney disease, 0.795 ± 0.007 for diabetes mellitus type 1, and 0.747 ± 0.006 for multiple sclerosis. Across all disease types, RETFound-based models outperformed the Xception CNN model, and notably, the OCT RLH model exceeded the performance of the CFP RLH model for both chronic kidney disease and multiple sclerosis.

TABLE 7. Area under the receiver operating characteristic curve (AUROC) for each of the models trained using 4,800 participants across the three selected diseases of interest.

Disease type	Xception (CFP)	RLH (CFP)	RLH (OCT)	RLH (CFP + OCT)
Chronic kidney disease	0.624 ± 0.024	0.637 ± 0.003	0.678 ± 0.003	0.683 ± 0.003
Diabetes mellitus type 1	0.707 ± 0.023	0.757 ± 0.006	0.729 ± 0.005	0.795 ± 0.007
Multiple sclerosis	0.651 ± 0.029	0.706 ± 0.004	0.732 ± 0.004	0.747 ± 0.006

D. COMPUTATIONAL EFFICIENCY

On average, the multimodal CFP + OCT RLH model required 5.0 TFLOPs per participant for training. In comparison, the single-modality CFP RLH model required 0.2 TFLOPs per participant, and the single-modality OCT RLH model required 4.8 TFLOPs per participant. In contrast, the Xception model required 33.0 TFLOPs per participant. Therefore, the proposed multimodal RLH model is about 6.6 times more computationally efficient to train than the Xception model, even though the Xception model processes only 2D CFP data.

IV. DISCUSSION

In this work, it was shown that the proposed multimodal RLH model, which combines high-level imaging features extracted from OCT and CFP images, achieved a new state-of-the-art accuracy for retinal age prediction with an MAE of 2.75 years when trained using data from 4,800 UK Biobank participants. Furthermore, experimental results demonstrated that the proposed multimodal model surpassed the performance of RLH

models trained solely on CFP images (MAE of 3.21 years) or OCT images (MAE of 3.98 years), and the commonly used Xception model trained on CFP images (MAE of 3.36 years). Moreover, the multimodal RLH model exceeds the accuracy of previously reported results from the biological retinal age estimation literature (although not directly comparable). For example, Chen et al. reported an MAE of 3.27 years using OCT images [24] and Zhu et al. achieved an MAE of 3.55 years for CFP data [6].

Importantly, the classification results indicate that multimodal RAG estimates hold significant promise for detecting non-ocular diseases, consistently outperforming single-modality models as measured by AUROC scores. To the best of our knowledge, this study is the first to demonstrate that enhanced accuracy in retinal age estimation, achieved by integrating CFP and OCT modalities, can lead to improved downstream capabilities in disease detection. In addition, RETFound-based models achieved higher AUROC scores than the commonly employed Xception CNN model, underscoring the effectiveness of foundation model-driven feature extraction for retinal age prediction. Notably, the multimodal CFP + OCT RLH model surpassed the CFP RLH model in classifying chronic kidney disease and multiple sclerosis, suggesting that OCT-derived structural biomarkers capture additional aging-related retinal changes associated with these conditions. This finding aligns with previous studies indicating that retinal layer thinning, choroidal alterations, and microvascular abnormalities that can be detected via OCT may serve as early indicators of systemic diseases [1], [3], [5], [6], [7], [17]. These results underscore the clinical potential of multimodal RAG analysis as a non-invasive screening tool for disease detection. By enhancing disease classification accuracy, multimodal RAG predictions could support more precise patient monitoring, facilitate earlier interventions, and contribute to the development of personalized healthcare strategies. Furthermore, the proposed framework can be employed in settings where only one of the modalities is accessible. This flexibility supports broader accessibility and implementation, especially in lower socioeconomic regions where OCT devices may not be widely available.

To further contextualize our findings, the proposed multimodal model (CFP + OCT) achieved an inter-eye MAD of 1.94 ± 0.03 years between estimated retinal ages of the left and right eyes, the lowest inter-eye variability among all evaluated models. In particular, our RETFound-based model demonstrated significantly greater inter-eye consistency compared to the widely used Xception model, which yielded a MAD of 2.72 years. Notably, the Xception architecture has been extensively employed in previous RAG studies and has shown strong potential as a screening biomarker for kidney failure [1], stroke [2], cardiovascular disease [7], and metabolic disorders [18]. Thus, by achieving improved inter-eye agreement, our model enhances the stability and reliability of retinal age estimation over an already clinically promising baseline. Furthermore, our

results compare favorably to reproducibility analyses such as Zoellin et al. [44], which reported inter-eye MADs exceeding 3.3 years under standardized conditions. It is worth noting that this degree of variability is not unprecedented among clinical biomarkers. Blood pressure (BP), for instance, one of the most routinely used metrics in medicine, exhibits considerable intra-individual variation due to diurnal cycles, physical activity, and stress, alongside inherent measurement noise [45], [46]. Despite these fluctuations, BP remains indispensable in clinical decision-making. This analogy underscores a broader point: biological or measurement-related variability does not preclude a biomarker from being both effective and actionable when interpreted within an appropriate clinical framework.

The improved retinal age prediction capability of the proposed RLH model further underscores the value of utilizing foundation model-based feature extraction in combination with multimodal input data. Additionally, the linear structure of the multimodal regression head model allows for the relative importance of CFP and OCT-derived features to be easily measured, thereby enhancing model interpretability [47]. Furthermore, the performance advantage of the proposed RLH model is even more pronounced with smaller datasets, showing its suitability for limited data scenarios. Notably, for this research, the CFP-RETFound and OCT-RETFound models used for feature extraction were employed directly as publicly released without undergoing any further finetuning or pretraining. Consequently, the training demands for the multimodal RLH model, were shown to be approximately 6.6 times lower in terms of FLOPs than the commonly used Xception model, even though the multimodal RLH model processes 2D CFP and 3D OCT data both, compared to the Xception model's reliance on only 2D CFP data only. The improved training efficiency makes it potentially feasible to integrate this method seamlessly and efficiently within a federated learning framework for use on mobile devices [48]. This is particularly vital for promoting global access to specialized and adapted retinal age prediction models, especially in regions with limited computational resources.

In this work, the training data consisted exclusively of healthy individuals, whereas RAG estimates were calculated for individuals with various diseases. Consequently, the data from individuals with diseases likely fall outside the training distribution of the model, essentially making them so-called out-of-distribution samples. The model's ability to generalize to these samples is crucial for RAG reliability. Our results, summarized in Table 6, highlight the multimodal RLH model's consistent extrapolation ability beyond the training data, demonstrating more consistent RAG estimation performance for individuals with diseases. Additionally, the observed reduction in MAD with increasing training data further highlights the robustness and scalability of the multimodal model. Unlike the Xception-based single-modality approach, which did not show a decrease in MAD, the RETFound-based models benefited from larger training datasets, leading to more consistent retinal age predictions.

This suggests that foundation model-based multimodal learning effectively captures shared aging patterns across both eyes, improving prediction reliability and potentially enhancing its applicability in clinical settings.

The RAG, as evaluated in this study, is intended to function as a general-purpose, non-invasive screening biomarker for the presence of many diseases rather than a disease-specific diagnostic tool – similar to many blood tests. Rather than classifying individual diseases directly, the RAG captures deviations from expected healthy retinal aging and can help flagging individuals who may benefit from further clinical assessment and more detailed clinical investigations. This is conceptually similar to other early risk indicators, such as elevated C-reactive protein in blood samples, which highlight systemic irregularities without pinpointing a specific condition [49]. A key motivation for focusing on retinal age prediction rather than direct disease classification lies in the scalability and flexibility of the approach. Disease-specific classifiers require large, well-annotated datasets for each target condition, which represent resources that are often limited or unavailable in many real-world settings. This challenge is compounded by the fact that many individuals present with multiple concurrent health issues (comorbidities), creating complex multi-label classification scenarios that are inherently difficult for machine learning models to solve and significantly complicate the process of acquiring accurate, distinct labels. In contrast, retinal age models can be trained on large datasets from healthy individuals requiring only the chronological age as an input aside from the imaging data, a universally available and low-cost label. This enables the RAG to act as a generalized biomarker of systemic health, applicable across a broad spectrum of diseases, potentially including those that may be underdiagnosed or inconsistently labeled. Thus, the RAG offers a complementary role within clinical workflows, serving as an early indicator to inform prioritization, rather than replacing condition-specific diagnostic tools.

The potential of the RAG as a disease biomarker directly depends on its ability to generate higher scores for patients with diseases than for healthy controls. Consequently, it is anticipated that systemic and inflammatory diseases such as diabetes mellitus type 1, multiple sclerosis, and chronic kidney disease, which are all associated with identifiable retinal manifestations, should yield higher RAG scores. For instance, diabetes mellitus type 1 can present with pathological features such as microaneurysms, hemorrhages, hard exudates, and neovascularization [50]. Similarly, chronic kidney disease can lead to changes in the retinal vasculature and thickness [51], and multiple sclerosis may result in noticeable alterations of the optic disc and retinal nerve fiber layer thickness [52]. Our results demonstrate that the RAG estimates from RLH models not only surpass those from the Xception model but also tend to improve with more training data, enhancing the model's ability to detect disease-related anomalies. Notably, RAG scores derived from the RLH model trained on OCT data were higher for multiple sclerosis

and chronic kidney disease than those trained on CFP, while scores for diabetes mellitus type 1 were comparable. This is physiologically viable, as nerve fiber layer thickness and changes in retinal thickness and volume can be more easily observed in OCT data than CFP data [53]. Conversely, the physiological alterations associated with retinopathy from type 1 diabetes mellitus are consistently observable in both OCT and CFP data, explaining the similar RAG scores across these imaging modalities [54]. These observations indicate that different imaging modalities may reveal varying levels of information about these diseases, highlighting the importance of incorporating both modalities into model training and for RAG calculation.

While this work provides a comprehensive examination of the proposed multimodal RLH model for predicting retinal age, it is important to acknowledge certain limitations. For instance, the data used in this research was sourced exclusively from a UK population. Thus, expanding the dataset to other populations would be advantageous to better understand the applicability of the findings on a global scale. Furthermore, a substantial number of images from the UK Biobank were excluded from analysis due to failing quality control screening. This observation is consistent with findings by MacGillivray et al. [55], who reported that a considerable proportion of UK Biobank retinal images were not suitable for automated processing due to quality limitations. However, in practical clinical settings, image quality is generally evaluated at the time of capture or immediately thereafter. This allows for the identification of suboptimal images, which can be promptly retaken, thereby minimizing the impact of poor image quality on downstream analyses as proposed in this work. Additionally, future efforts should focus on a prospective multi-center clinical validation to test the multimodal model in diverse healthcare settings, ensuring robustness across different clinical workflows, imaging protocols, and image resolution and quality levels. Moreover, there is a urgent need to evaluate RAG estimation performance using data from portable or lower-cost imaging systems, such as smartphone-based fundus cameras [56] or emerging at-home OCT devices to further assess real-world generalizability [57]. Additionally, this research focused solely on comparing the RLH models with the Xception CNN architecture. Despite the rationale for selecting the Xception architecture due to its prevalent use in the retinal age prediction literature [1], [2], [3], [5], [6], [7], further comparative studies involving other architectures would support the generalizability and superiority of the proposed method. Furthermore, while our model significantly improves inter-eye prediction stability compared to previous work, further reducing this variability is a key direction for future research. Potential avenues include developing distinct left-eye and right-eye models, implementing prediction averaging across both eyes, or employing advanced architectures such as Siamese networks [58], specifically designed to minimize inter-eye output differences during training. Finally, it should be highlighted that while the term “retinal age gap”

has been widely adopted in the literature to describe the discrepancy between predicted biological age from ocular imaging and chronological age [59], we acknowledge that this terminology may imply a more anatomically restricted signal than the imaging data truly provides. Specifically, the OCT volumes used in this study were acquired under standard clinical conditions and include not only the retina, but also adjacent ocular structures such as the vitreous and choroid [60]. Similarly, CFP images may capture extra-retinal features, including the choroidal pigmentation [61]. Therefore, the predicted biological age in OCT-based or multimodal (CFP + OCT) models should be interpreted more broadly as an ocular-image-derived estimate of systemic aging, rather than one that stems exclusively from the retina. This reframing does not diminish the clinical or research utility of the RAG but rather broadens its biological interpretation to better reflect the diverse anatomical compartments contributing to the model's predictions.

V. CONCLUSION

In summary, this work demonstrates the potential of the RAG estimated using multimodal imaging as a clinically relevant, non-invasive imaging biomarker capable of identifying individuals who may benefit from further, more targeted diagnostic evaluation. The enhanced disease classification accuracy achieved through multimodal integration strengthens the case for incorporating multimodal RAG assessment into routine clinical workflows, where it can systematically screen for a wide range of diseases. This capability has the potential to enable earlier disease detection and improve long-term patient outcomes.

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